CSE-115
CONTROL STATEMENTS (IF-ELSE, SWITCH CASE, NESTED IF-ELSE)

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Week 4 Evaluation

Make a fine calculator for absent students that takes the following inputs and gives outputs accordingly.
Follow the output format closely and declare variables in such a way that minimal space in used.
Note that, fine = BDT 50 * absent days;

```
                Attendance Fine Calculator


Enter number of school days last month: 30

Enter Roll of student: 202236001

Enter number of days student was present: 25


202236001 was absent in 5 out of 30 days last month
Attendance percentage: 16.67%, FINE: 250


Process returned 0 (0x0)   execution time : 17.876 s
Press any key to continue.
```

FOR ALL SOLVED CODES, CLICK HERE

TYPES OF CONTROL STATEMENTS

Type	When to Use	Example Case
A single If	Just need to check one condition	Bill checking software. You only except positive bill. Check if the input is >0.
Sequence of only "If"s	Need to check multiple condition one after another (usually on same vars, might be true for multiple cases)	Check seq_of_ifs.c
If and else	Only two conditions, only one of them might be true. Second condition doesn't even need to be checked.	Simple pass fail calculator. If number>=33 passed. Otherwise Failed.
If- else-if- else Ladder	Only one condition out of multiple may be true. Check them one after another. The last or default condition is not checked and written with an "else"	Check if_else_grade.c

TYPES OF CONTROL STATEMENTS

Type	When to Use	Example Case
Switch Case	Exact same as If-Else if-Else. Just different syntax. Matter of preference.	For example, menu for any program. (Press 1 to Add, 2 to divide)
Ternary Operator	Used instead of plain if-else usually. Can also be nested to replicate if-else if- else Ladder.	Sim to if-else if_else programs. Check out here.
Nested If-else	If one condition is true, I might have to check another bunch of conditions to find out the ultimate result.	Check MIST_course_picker.c or coordinate_selection.c

SLIDE 6-20 IS FOR RECAP ONLY.
WE ALREADY COVERED THEM IN THEORY.

ONLY "IF"

Syntax – Only "If"

-Practical example:

if it is not raining, we will go to MIST.

-Format:

if(check if it's raining)

printf("Let's not go to MIST!")';

-Format:

if(expression)

No Semicolon

statement;

Syntax – Multi Statement If

- Format:

```
if(check if it's raining)
{
        statement1;
        statement2;
        …………………..;
         statement n;
}
```

Syntax – Only “If” – An example

```
#include<stdio.h>
void main()
{
        int var;
        scanf(“ %d”, &var);
        if(var > 0)
                printf(“%d is a Positive Number”, var);
}
```

Syntax – Multi line "If" – An example

```
#include<stdio.h>
void main()
{
        int var;
        scanf(" %d", &var);
        if(var > 0)
        {
                printf("The number is a Positive Number");
                printf("\nThe number is: %d", var);
        }
}
```

SEQ OF "IF"S

Syntax – Seq of “If”s – An example

```
#include<stdio.h>
void main()
{
        int var;
        scanf(“ %d”, &var);
        if(var > 0)
                printf(“%d is a Positive Number”, var);

        if(var < 0)
                printf(“%d is a Negative Number”, var);
}
```

Syntax – Seq of If Ladder – An example

Check out seq_of_ifs.c

IF ELSE

Syntax – If-Else

-Practical example:

if **x** is divisible by 2, then x is an even number.

otherwise **x** an odd number

-Format:

```
if(expression) statement1;
else statement2;
```

-Format:

```
if(expression)              No Semicolon
        statement1;
else                        No Semicolon
        statement2;
```

SAMPLE PROGRAM - IF ELSE

```
#include<stdio.h>
void main()
{
        int var;
        scanf(" %d", &var);
        if(var > -1)
                printf("%d is a Positive Number", var);
        else
                printf("%d is a Negative Number", var);
}
```

CONTROL STATEMENT (CONT.)
IF ELSE

- Format:

```
if(expression)
{
        statement1;
        statement2;
        .....................
}
else
        statement 3;
```

CONTROL STATEMENT (CONT.)
IF ELSE

- Format:

```
if(expression)
        statement1;
else
{
        statement2;
        .....................
        statement n;
}
```

CONTROL STATEMENT (CONT.)
IF ELSE

- Format:

```
if(expression)
{           statement1;
             statement2;
            ……………………
}
else
{
             statement4;
            ……………………
            statement n;
}
```

CONTROL STATEMENT (CONT.)
IF ELSE

- Format:

```
if(expression)
{
        statement 1;
}
statement m;                    //Wrong
else                            //mismatched if else
{
        statement4;
        ......................
        statement n;
}
```

EXERCISE 1 – MULTI LINE IF-ELSE

Problem:

Take buying price and selling price of a product as input and find out the percentage of the profit or loss of the product.

Sample Input 1:

Buy: 10

Sell: 15

Sample Output 1:

Profit 50%

Sample Input 2:

Buy: 40

Sell: 20

Sample Output 2:

Loss 50%

Solution: [check out profit_loss.c]

```
#include<stdio.h>

int main(){
    double buying_price, selling_price,
        profit, loss;

    printf("insert buying price \n");
    scanf("%lf", &buying_price);
    printf("insert selling price \n");
    scanf("%lf", &selling_price);

if(buying_price < selling_price){
        profit= selling_price - buying_price;
        printf("Profit %lf", profit/buying_price*100);
    }
    else{
        loss = buying_price - selling_price;
        printf("Loss %lf", loss/buying_price*100);
    }
}
```

LOGICAL OPERATORS – COMBINE CONDITIONS

Symbol	Operator_Name
&&	AND
||	OR
!	NOT

EXERCISE 2

Problem:

Take a character as input and determine whether it is a vowel or not.

Sample Input 1:

A

Sample Output 1:

Vowel

Sample Input 2:

B

Sample Output 2:

Consonant

Solution: check out vowel.c

EXERCISE 3

Problem:

Take a year as input and find out whether it is leap year (check the leap year algorithm) or not.

Sample Input 1:

2012

Sample Output 1:

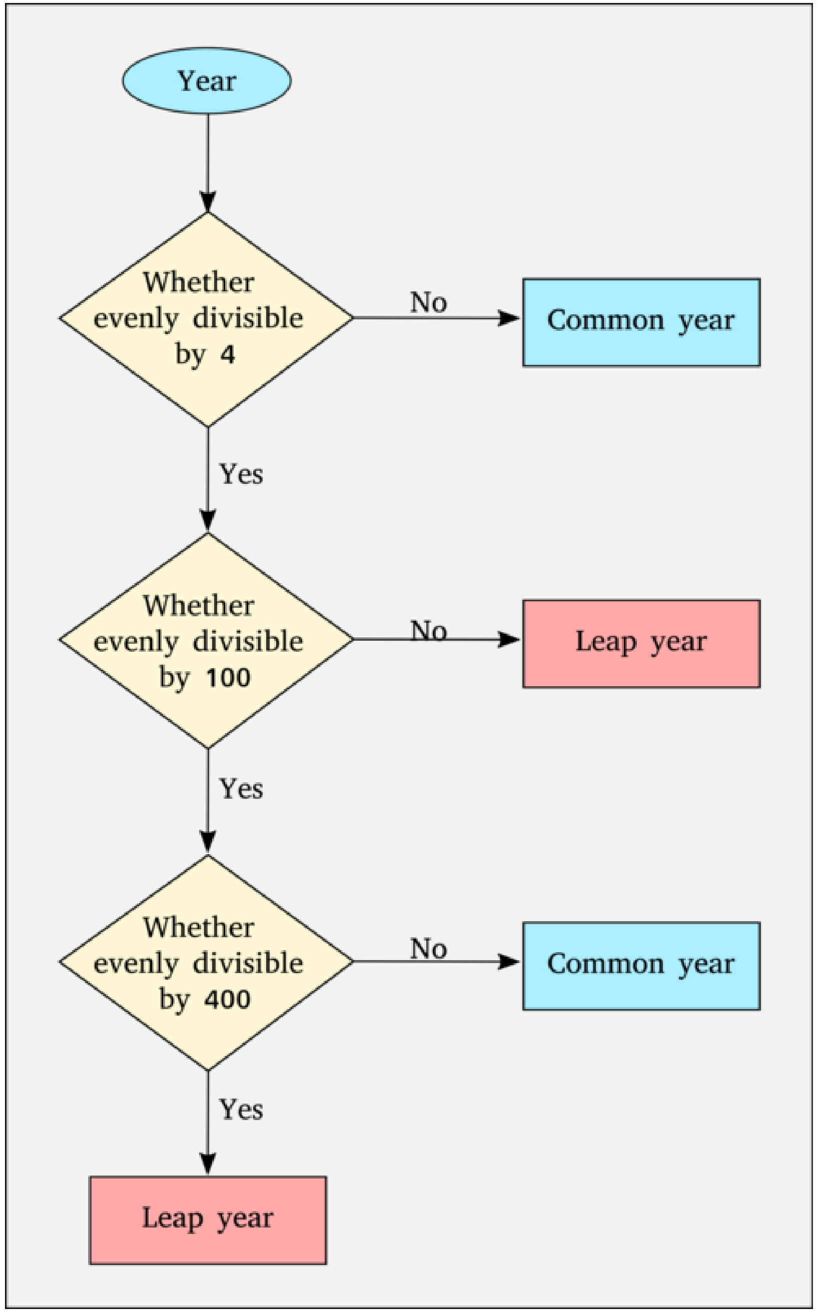
Leap Year

Sample Input 2:

2010

Sample Output 2:

Not Leap Year



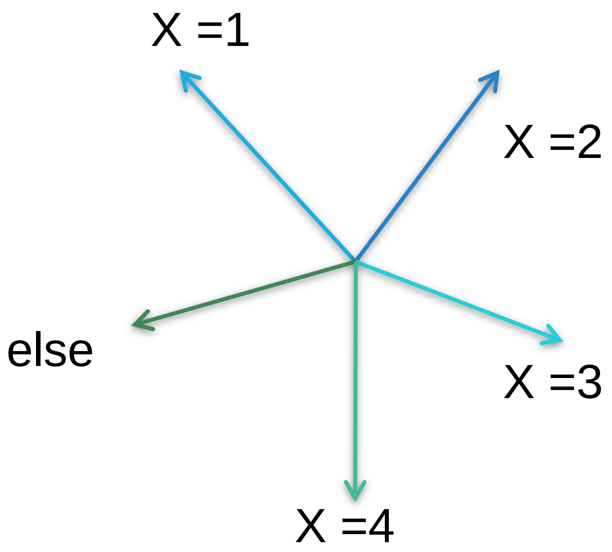
Solution: check out if_else_leap_year.c

"IF-ELSE IF-ELSE" LADDER

ELSE IF LADDER

- Format:

```
if(expression)
        statement1;
else if(expression)
        statement2;
..........................
..........................
else if(expression)
        statement_n;
else
        statement _x;
```



EXERCISE 4:

Take one student's number as input and find out the grade of the mentioned student.

Number	Grade
80-100	A+
75-79	A
70-74	A-
65-69	B+
60-64	B
55-59	B-
50-54	C+
45-49	C
40-44	D
<40	F

Sample Input 1:
Number: 76
Sample Output 1:
Grade: A

Sample Input 2:
Number: 44
Sample Output 1:
Grade: D

Solution: check out if-else_grade.c

```
#include<stdio.h>
void main()
{     int a;
      scanf(" %d", &a);
      if(a >= 80)
            printf("A+");
      else if(a >= 75 && a<80)
            printf("A");
      else if(a >= 70 && a<75)
            printf("A-");
      else if(a >= 65 && a<70)
            printf("B+");
      else if(a >= 60 && a<65)
            printf("B");
      else if(a >= 55 && a<60)
            printf("B-");

      ...................................
      ...
      ...................................
      ...

      else if(a >= 40 && a<45)
            printf("D");
      else
            printf("F");
}
```

SWITCH CASE

EXERCISE 5 – FOR SWITCH CASE

Problem:

Write a program that takes two integer numbers and print the following menu

1. Addition
2. Subtraction
3. Multiplication
4. Division

Now take the user choice i.e. 1, 2 etc and print the result.

Screen Output:

Insert two numbers: 21 11
1. Addition
2. Subtraction
3. Multiplication
4. Division
Choice: 2
Output: 10

SWITCH CASE

```
switch(expresssion)
{       case   constant_1:
                        statement1;
                        break;
        case   constant_2:
                        statement2;
                        statement3;
                        break;
        ................................
        default:
                        statement_default;
                        break;
}
```

Example 1: Switch_age.c
Example 2: Switch_char.c

Sample Program

```
#include<stdio.h>
void main()
{
    int x, y, r, ch;

    printf("1.Add    2.Sub");

    scanf(" %d", &ch);

    scanf(" %d %d", &x, &y);

    switch(ch)
    {
        case 1 :
            r = x + y;
            printf("%d", r);
            break;
        case 2 :
            r = x - y;
            printf("%d", r);
            break;
    }
}
```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Sample Program (cont.)

```
#include<stdio.h>
void main()
{
        int x, y, r, ch;

        printf("3.Add     4.Sub");

        scanf(" %d", &ch);

        scanf(" %d %d", &x, &y);

switch(ch)
{
        case 3 :
                r = x + y;
                printf("%d", r);
                break;
        case 4 :
                r = x - y;
                printf("%d", r);
                break;
        default:
        printf("wrong choice");
        break;
}
}
```

Sample Program (cont.)

```
#include<stdio.h>
void main()
{
        int x, y, r;
        char ch;
        printf(“a.Add     s.Sub”);

        scanf(“ %c”, &ch);

        scanf(“ %d %d”, &x, &y);
```

```
switch(ch)
{
        case ‘a’ :
        case ‘A’ :
                r = x + y;
                printf("%d",r);
                break;
        case ‘s’ :
        case ‘S’ :
                r = x - y;
                printf("%d",r);
                break;
}
}
```

EXERCISE 6: - Solve With Switch Case:
Write a C program to take input of a character and three integers and make the following menu. Take the inputs as double or float.

Character	**Output (Values)**
A	Sqrt(x) + y^4 + 6z
B	(x % y) / z
C	Corresponding characters of x, y and z
Others	Invalid Input

*Note: Use sqrt() and pow() in math.h to execute Case A

Sample Input	Sample Output
A 2 2 3	Output: 35.41
C 40 90 49	Output: (Z 1
B 11 4 2	Output: 1.50

Solution: check out switch_case_prob.c

NESTED IF-ELSE

NESTED IF-ELSE

- Format:

```
if (expression1)
{
    if(expression2)
        statement1;
    else
        statement2;
}
```


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```
if (expression1)
{
    if(expression2)
        statement1;
    else
        statement2;
}
else
{
    if(expression3)
        statement3;
}
```


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```
#include<stdio.h>

void main()
{     int  x;
      scanf(" %d", &x)
      if(x  > 5)
      {
          if(x  <  10)
               printf("%d is within 5 and 10", x);
      }
}
```


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```
#include<stdio.h>

void main()
{      int  x;
       scanf(" %d", &x)
       if(x  > 5)
       {    if(x  <  10)
                  printf("%d is within 5 and 10", x);
       }
       else
                   printf("%d is not within 5 and 10", x);
}
```


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```
#include<stdio.h>

void main()
{       int  x;
        scanf(" %d", &x)
        if(x  > 5)
              if(x  <  10)
                     printf("%d is within 5 and 10", x);
              else
                     printf("%d is greater than 5 ", x);
}
```


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```
#include<stdio.h>

void main()
{       int  x;
        scanf(“ %d”, &x)
        if(x  > 5 && x < 10)
        {
                printf(“%d is within 5 and 10”, x);
        }
        else
                printf(“%d is not within 5 and 10”, x);
}
```


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EXERCISE 7

Question: MIST Course Picker

Solution: check out MIST_course_picker.c

EXERCISE 8

Problem:

Write a C program to accept a coordinate point in a XY coordinate system and determine in which quadrant the coordinate point lies. Also find out if the point is on the x or y axis. Or if the point is origin/center point itself!

Sample Input

7 9

-1 0

Sample Output

The coordinate point (7,9) lies in the First quadrant.

The coordinate point (-1,0) is on the X axis.

Solution: check out coordinate_solution.c

EXERCISE 9

Note: Do On your own. No need to submit. Will randomly ask a few students to share screen and show the code in next class. For use of getchar() before scanf() while taking char input, refer to the video at time mark 01h:50m:49s

Problem:

Write a C program to take a character. Find out whether the person is a **student** or an **employee.** If he is a **student** then take his **earned credit** as input and find out **which level** he belongs to.

If he is an **employee** then take **another character** as input and find out whether he is **Military** or **Civil** Faculty. If **civil faculty** then again **take another character** as input and find out whether he is **permanent** or **adjunct.**

Following chart has been given for better understanding.

EXERCISE

Character	Earned Credit	Output	
S	0-36	Level 1	
	36-72	Level 2	
	72-108	Level 3	
	>108	Level 4	
Character	Faculty	Output	Output
E	M	Military Faculty	
	C	Civil Faculty	
		Input	
		P	Permanent
		A	Adjunct

Solution: check out nested_conditional_logic_practice_problem.c

Sample Evaluation Questions

THANK YOU